Response to Reviewer Comments

Manuscript: "Machine learning in battery management systems: a structured scoping review and classification of 103 peer-reviewed studies, 2019–2026" (extended version)

We thank the reviewer for a searching and, we believe, correct critique. The central charge — that a quota-curated sample was being read as if it measured the field — is accurate, and fixing it required structural changes rather than wording. Below we respond point by point; all quoted locations refer to the revised manuscript. Every changed number re-verifies against the released dataset (consistency_check.py).

A note on scope, stated plainly

This revision is substantially rescoped relative to the PDF version previously seen. (i) The Tier-1 "measured bibliometrics" arm has been dropped as a measurement claim: the 799-record pool is now used only for explicitly caveated, quota-free baselines (function-tag counts whose tagger is validated against the classified subset), and its retrieval-shaped year mix is disclosed as such. (ii) The standards mapping has been folded into the reporting checklist (Table 11) and the validation-standards discussion (Section 9.1). (iii) The dedicated transportation-electrification section has been folded into the conference sensitivity set of Section 3.5. We state this openly because a reader of both versions will otherwise experience the revision as a different paper; we believe the rescoping is required by Reviewer point 1, since the earlier version's measured-bibliometrics claims exceeded what a quota-curated sample can support.

1. Quota-driven bibliometrics (circularity)

Fixed by re-anchoring and by deletion, not by defense. Section 2.3 now states the within-stratum selection rule explicitly: within each function stratum, records were ranked by Crossref citation count and taken from the top, with the ranking walked deeper to cover recent years — deterministic, not random, not citation-blind. The 98 + 5 decomposition (98 drawn from the pool, 5 individually retrieved seeds) is now stated in the text and in Fig. 2. The abstract and conclusion no longer claim a "clear five-year evolution" or any field-level trend; Section 3.1 now states that neither the subset (quota-balanced by construction) nor the pool (retrieval windows of unequal year coverage, Section 2.2) measures field output, and the age-normalization claim has been deleted. Where pool-level measurement is possible it is now used: Section 3.3 reports function-tag baselines from the 799-record pool (SOH 158, SOC 133, RUL 103, thermal 99, fault 96, charging 64, balancing 38), with the tagger validated against the subset's full-text labels (53 of 75 core-area papers, 71% any-tag agreement). Consequences we accepted: "thermal under-served" is withdrawn (99 pool tags, comparable to fault and safety; the real deficit is that no subset paper closes the thermal control loop on hardware, an evidential claim, Section 9.6), while "balancing under-served" now rests on pool-level evidence (smallest core function at both levels) rather than on quota-set counts. Table 10's evidence column is revised accordingly. What we did not do: hand-verify function tags for all 799 records; the pool baselines are aggregate-ordering statements with the measured tagger error disclosed, and Section 2.3 says exactly what they are.

2. Reliability study

All four requested items are addressed; one required an admission rather than new data. (i) Coder identity is disclosed: coders B and C are co-authors S.S.I.A. and J.I.C.; the blinding was procedural (locked, hash-tracked sheets; filed blinding memo; no access to released labels), not external, and their later author-contribution role in verifying classifications and counts is stated. (ii) The $\kappa = 0.48$ result is now scoped as what it is: abstract-level reproducibility (coder B had 7 full texts, coder C 2, of 30). (iii) Bootstrap 95% CIs are reported for all five fields and the strong/non-strong bucket ($\kappa = 0.52$, CI 0.09–0.84), together with the coders-versus-released-labels bucket kappas (0.22 and 0.42) and the pre-declared conservative weak-code fallback that drives them.

(iv) The noise model is written out in Section 3.4: per draw, a flip rate ~ Normal(mean 0.35, sd 0.02, truncated at 0) from the observed coder-versus-label bucket error rates (0.40, 0.30); symmetric strong/weak flips across the 88 primary papers; 4,000 seeded draws (seed 7). Finally, the admission: no full-text dual coding was performed. The manuscript now says so explicitly and states that the strong/weak counts rest on a single full-text coding by the lead author, buffered by the sensitivity analysis and by the full release of labels, sheets, and itemized disagreements. Full-text dual coding of at least 30 papers remains the correct next step; it requires coder time we have not yet invested, and we prefer to say that than to relabel what exists.

3. "Strong protocol" definitions

Restored. Section 2.5 now defines the three stringency tiers in place: strong (protocol-only) = cross-cell, cross-chemistry, cross-domain holdout, and online/real-vehicle demonstration; weak = within-cell splits and k-fold without cell-level isolation; environment-only = experimental hardware, simulation, and case study, with the broader definition additionally counting experimental-hardware demonstrations as strong. Sections 3.4 and the abstract now reference these definitions by name.

4. Fig. 2 contradictions

Regenerated to match Section 2.3. The figure no longer says "Two-tier synthesis: measured bibliometrics (Tier 1)" or "no quotas" in a way that contradicts the quota-based curation; it now shows the actual funnel: 799 rule-based pool → stratified draw of 107 (citation-ranked within strata) + 5 seeds = 112 → full-text curation (−19, +10) → 103 classified (98 pool + 5 seeds). The 98 figure is in the text. The clipped top box is fixed (the axes now enclose all boxes).

5. Claims that survived without evidence

(i) The conclusion no longer claims accuracy "progress": it now states that the volume of published accuracy claims is not matched by validation rigor — a count-based asymmetry — consistent with Section 5.6's finding that accuracy metrics cannot be pooled. (ii) "Largely solved on laboratory terms" (4.3) and "largely succeeded" (conclusion) are reworded to "report laboratory-regime success, on metrics that cannot be pooled across papers." (iii) Section 9.1 no longer asserts undisclosed splits, seeds, and compute budgets as counted fact; it states a qualitative observation and points to Section 9.7's disclosure that only validation, data-source, and artifact items were score-measured. Scoring all 103 studies against the 12 checklist items would indeed be cheap and is an excellent suggestion for the released-dataset community; doing it credibly requires the full texts, which the review team did not retain for all 103 studies, and we chose honesty about that over a retro-fitted audit.

6. The LLM characterization (ref [15], HA-LLM)

Checked against the source (ScienceDirect record for 10.1016/j.ensm.2026.104983). The reviewer is right: HA-LLM is an LLM-based SOH estimator — charging-current sequences aligned to the LLM embedding space, degradation priors injected via prefix adaptation, a Monte-Carlo uncertainty module that selectively substitutes a physics-consistent estimator, validated on one year of field data from 100 production vehicles with teardown/bench calibration on an unseen vehicle model. Section 8.4 is rewritten accordingly: the corpus's two LLM studies occupy different seats (incident-risk reasoning vs. uncertainty-gated estimation), and the estimator-seat study is presented as evidence that the guardrails this review proposes (uncertainty gating, physics fallback) are already being implemented in practice. Section 9.5, Table 2, and Table 9 now agree with this reading and with each other.

7. Reproducibility

The Zenodo DOI is restored as the deposit of record (10.5281/zenodo.23436202); the GitHub link is repositioned as a development mirror — which also cures our own checklist item 12. The data- and code-availability statements now enumerate every artifact promised in Sections 2.5 and 3.5: the classified 103-study dataset, the 799-record pool with query-theme tags, the screening log, the protocol audit, the codebook and coding protocol, the locked coder sheets and agreement/dispute reports, the conference sensitivity set, the withheld reliability labels, the sensitivity script and outputs, the Crossref audit trail, the seed register, and the statistics — plus all regeneration scripts.

8. AI disclosure

Merged into one statement. The Declarations entry now names each tool and its role: an AI language model (GLM, Z.ai) for literature-pooling, analysis, and figure-generation scripts; OpenAI ChatGPT, Grammarly, and QuillBot for language editing. It states explicitly that no AI tool decided study inclusion and no AI tool assigned or altered any classification label, that Section 2.5 documents who coded what, and that the coding evidence is released. The separate Acknowledgment listing the same tools for "language only" has been removed. (Authors: please verify this statement matches the facts of your process before submission.)

Smaller items

Abstract reduced from ~360 to ~300 words; "a much-anticipated finding is checked rather than assumed" removed; the function list now uses one basis (all-corpus totals). The age-normalized post-2023 claim is deleted (3.1). The venue "impact in application terms" inference is replaced with the venue counts themselves (7 of 103 IEEE; 9.0% of the pool) and a Crossref-registry caveat. The physics-informed monotonicity claim is scoped to the subset with the partial 2026 year flagged, and a pool-level cross-check (no monotonic tag trend once retrieval windows are accounted) is added. The 12-review cap versus 15 review-labelled papers is explained in Section 2.4 (12 reviews + 3 review-type syntheses coded under their dominant method family).

We believe the manuscript is now internally consistent: every distributional claim states its evidentiary level (subset composition, pool baseline, or qualitative reading), the reliability and sensitivity machinery is fully specified, and the LLM characterization matches the sources. We are grateful for the review, which materially improved the paper’s honesty about what a curated corpus can show.